Nishanth Ravula

Software and ML Engineer

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Summary

Software and ML engineer who builds anomaly detection and data systems for retail operations. Experience across production data infrastructure, deep learning for autonomous-vehicle perception and production web front ends. PhD student in AI researching self-supervised time-series anomaly detection, with six published audits of AI systems for fairness and regulatory compliance.

Experience

Software Developer, Dollar General April 2024 to present

- Build anomaly detection for retail ordering data across the store network.
- Tune and reconcile Oracle workloads behind the ordering pipeline, including query rewrites for correctness and speed.
- Prototyped a DuckDB and FastAPI analytical layer that serves dashboards from versioned snapshots, taking load off production databases; validated with async load tests, Locust and per-request server timing.
- Introduced JSON-schema validation for MongoDB inputs, reducing user-side data errors by 40%.
- Keep Oracle and MongoDB in sync and maintain the CI/CD pipelines that ship them.

Assistant System Engineer, Tata Consultancy Services July 2021 to July 2022

- Built and trained deep learning models for autonomous-vehicle perception in TensorFlow.
- Improved model accuracy by 20% through better training data, regularization and hyperparameter tuning.
- Led the data annotation team and designed the quality-control process behind the training set.
- Deployed models and managed the ML lifecycle on Azure ML.

Web Developer, Halfway February 2020 to June 2021

- Built the React and Next.js front end with Google OAuth and Maps APIs.
- Improved page performance by 35% with component-level optimization and code-splitting.

Frontend Developer Intern, PrimeFort October 2019 to January 2020

- Built React interfaces and reusable components for client websites.

Selected projects

RSTAD: self-supervised anomaly detection for retail time series Paper in preparation

Transformer masked autoencoder with dual-branch scoring: reconstruction error plus distance to a kNN memory bank of normal patterns, targeting the over-generalization failure of memory-based detectors. Benchmarked on SMD, SMAP, MSL and NAB. PyTorch, MLflow.

Responsible AI audits 6 published on GitHub

Audits of real and realistic AI systems against the EU AI Act, GDPR and NIST AI RMF. Applying equal opportunity cut a recidivism model's error-rate disparity by 85% for 0.63 points of accuracy.

mcp-guardrail Open source, TypeScript

A stdio proxy for the Model Context Protocol that redacts secrets and personal data from AI agents' tool results before they reach the model. CLI and test suite with CI.

Agent tooling on the Model Context Protocol TypeScript, React

A LinkedIn MCP server with OAuth 2.0 and confirm-before-post, and a job-application agent over the Indeed MCP server that tailors résumés and drafts cover letters.

Education

PhD, Artificial Intelligence, University of the Cumberlands In progress

MS, Computer Science and Engineering, University at Buffalo, SUNY 2022 to 2023

BTech, Computer Science and Engineering, Sri Indu College of Engineering and Technology 2017 to 2021

Skills

Machine learning	PyTorch, TensorFlow, scikit-learn, MLflow, Azure ML, self-supervised learning, anomaly detection
Data	Oracle, PL/SQL, DuckDB, Parquet, MongoDB, PostgreSQL, MySQL, Solr
Backend	Python, FastAPI, Flask, Django, Node.js, REST
Frontend	TypeScript, JavaScript, React, Next.js, Redux
AI agents	Model Context Protocol, OAuth 2.0, LLM evaluation
Delivery	Docker, Jenkins, CI/CD, Git, Locust, GCP, Azure
Governance	EU AI Act, GDPR, NIST AI RMF, fairness metrics, algorithmic impact assessment